

PAYAM PARVIZI

Robotics & Reinforcement Learning Research Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Authorized to work in Canada | Open to relocation across Ontario

SUMMARY

PhD-trained research engineer specializing in continuous-control reinforcement learning, custom physics-based simulation, policy smoothness, and sim-to-real systems. Implemented PPO, SAC, and DDPG from scratch in PyTorch; built MuJoCo/Gymnasium environments; and scaled multi-seed studies on SLURM clusters. Led research from mathematical method design through software, experiments, analysis, figures, and publication.

TECHNICAL SKILLS

Reinforcement Learning: PPO, SAC, DDPG, actor-critic methods, continuous control, policy regularization, reward design, sim-to-real evaluation

ML / Simulation: PyTorch, Gymnasium, MuJoCo, Tianshou, NumPy, pandas, Matplotlib, Weights & Biases; JAX for selected experiments

Programming / Computing: Python and OOP, MATLAB/Simulink, Bash, Linux/Ubuntu, HPC/SLURM, Git/GitHub repository management, LaTeX

Robotics / Control: Crazyflie, cflib, Crazyradio, Flow Deck, hands-on hardware integration; classical control and system dynamics (teaching/theory)

EXPERIENCE

Research Associate (Postdoctoral Fellow) | University of Ottawa | Ottawa, ON | Nov 2025 - Mar 2026

- Advanced SAPPs, a trajectory-local policy regularization method that balances smoothness and responsiveness in continuous-control RL.
- Led implementation, cross-domain experiments, analysis, visualization, and manuscript development across MuJoCo, adaptive optics, and quadcopter control.
- Ran multi-environment, multi-seed CPU/GPU studies using Linux, HPC resources, SLURM, and Weights & Biases.

Research Assistant | University of Ottawa | Ottawa, ON | Sep 2022 - Aug 2025

University appointment on a collaborative project with the National Research Council of Canada.

- Designed and released Adaptive Optics Gym, a configurable wavefront-sensorless adaptive-optics RL environment spanning turbulence, deformable-mirror control, optical propagation, focal-plane sensing, fiber coupling, and multiple reward/action formulations.
- Implemented PPO, SAC, and DDPG from scratch in PyTorch, including actor-critic networks, replay buffers, training/evaluation loops, experiment logging, and checkpointing.
- Led code development, experimental design and execution, analysis, figures, and primary writing for peer-reviewed RL/adaptive-optics studies; maintained the associated public repositories.
- Demonstrated higher optical coupling and lower action fluctuation than vanilla PPO and fixed regularization baselines across multiple simulated turbulence conditions.

Teaching Assistant | University of Ottawa | Ottawa, ON | 2019 - 2022

- Supported Control Systems (4 terms), System Dynamics (2 terms), and Biomedical System Dynamics (1 term), including feedback control, stability, state-space modelling, and MATLAB/Simulink analysis.
- Nominated for the University of Ottawa Excellence Award for Teaching Assistants, 2022.

Project Assistant | Middle East Technical University | Ankara, Turkey | Nov 2015 - Nov 2017

University appointment on a collaborative project with TÜBİTAK, Grant 114E274.

- Developed DMP-based robotic deburring skill models from haptic-device demonstrations and evaluated adaptable trajectories on a hexapod platform.
- Implemented particle swarm optimization for DMP parameter identification and developed force-trajectory DMP methods and experiments.
- Built a MATLAB neural-network model relating process forces/load to depth of cut, spindle speed, and feed rate; thesis experiments achieved 39 micrometres maximum and 27.9 micrometres mean flat-surface form error.

SELECTED RESEARCH PROJECTS

State-Adaptive Proportional Policy Smoothing (SAPPS) | github.com/payamparvizi/sapps-rl

- Created a loss-function regularization method that adapts policy smoothing locally along trajectories instead of applying a fixed global penalty; evaluated it across MuJoCo, adaptive optics, and Crazyflie control.
- Led nearly all code, experiments, analysis, figures, and manuscript preparation; TechRxiv preprint under revision following peer review.

Adaptive Optics Gym | github.com/payamparvizi/adaptive_optics_gym

- Authored and maintained the complete research environment, optical simulation stack, command-line workflows, and from-scratch PPO/SAC/DDPG implementations.
- Supported configurable atmosphere dynamics, observations, actions, rewards, deformable-mirror control, fiber coupling, training, evaluation, logging, and checkpoints.

Crazyflie Reinforcement Learning | github.com/payamparvizi/Crazyflie_RL

- Built the simulation-to-hardware altitude-control workflow using PPO, Crazyradio, Flow Deck, cflib, safety constraints, automatic landing, and real-time telemetry.
- Completed 100 real-hardware episodes after multi-seed simulation training; inference was approximately 312 microseconds at a 10 Hz control rate.

Custom MuJoCo Truck-Packing Environment

- Implemented a Gymnasium-compatible simulator with truck/box geometry, contact physics, placement validation, stability checks, occupancy-grid observations, continuous actions, and density-based rewards.
- Trained a customized Tianshou PPO agent with a custom neural-network architecture; owned the simulator/environment and RL training component.

EDUCATION

PhD in Mechanical Engineering | University of Ottawa | Ottawa, ON | Sep 2018 - Dec 2025

Thesis: "Adaptive Policy Smoothing in Reinforcement Learning: Applications to Wavefront Sensorless Adaptive Optics and Robotics."

MSc in Mechanical Engineering | Middle East Technical University | Ankara, Turkey | Feb 2015 - Feb 2018

BSc in Aerospace Engineering | Middle East Technical University | Ankara, Turkey | Sep 2008 - Jun 2014

SELECTED PUBLICATIONS

- "Action-Regularized Reinforcement Learning for Adaptive Optics in Optical Satellite Communication," Journal of the Optical Society of America B, early posting, 2026. [doi:10.1364/JOSAB.578050](https://doi.org/10.1364/JOSAB.578050)
- "Adaptive Policy Regularization for Smooth Control in Reinforcement Learning," TechRxiv preprint, 2026; manuscript under revision. [doi:10.36227/techrxiv.177004949.91897305](https://doi.org/10.36227/techrxiv.177004949.91897305)
- "Reinforcement Learning Environment for Wavefront Sensorless Adaptive Optics in Single-Mode Fiber Coupled Optical Satellite Communications Downlinks," Photonics, 10(12), 1371, 2023. [doi:10.3390/photonics10121371](https://doi.org/10.3390/photonics10121371)
- "Parametrization of Robotic Deburring Process with Motor Skills from Motion Primitives of Human Skill Model," MMAR, pp. 373-378, 2017. [doi:10.1109/MMAR.2017.8046856](https://doi.org/10.1109/MMAR.2017.8046856)

RECOGNITION AND ADDITIONAL INFORMATION

Recognition: University of Ottawa Excellence Award for Teaching Assistants nominee, 2022; MITx verified certificates in Probability, Machine Learning with Python, and Fundamentals of Statistics. **Languages:** Persian (native/bilingual), Azerbaijani (native spoken), Turkish (full professional), English (full professional). Authorized to work in Canada; full Ontario G driver's licence.